Ruomeng Ding

ruomeng@cs.unc.edu Google Scholar GitHub Homepage

Research interests: LLM Agents & Reasoning · Preference Learning & Elicitation · Reliable LLM Evaluation · AI4AI

EDUCATION

Ph.D. in Computer Science, University of North Carolina at Chapel Hill — Aug 2025 – Present

- Advised by Dr. Zhun Deng.

M.S. in Computer Science, Georgia Institute of Technology — Aug 2022 – May 2025

- GPA: 4.00/4.00; specialization: Machine Learning.

B.Eng. in Computer Science and Technology, Tianjin University — Aug 2018 – May 2022

- GPA: 3.75/4.00; major rank: 11/169 (top 6.5%).

PUBLICATIONS & PREPRINTS

** Equal contribution. Workshop papers are identified explicitly.*

LLM Agents & Reasoning

1. [Preprint] [AutoMLR Workshop @NeurIPS 2026] **R. Ding**, Q. Shen, Z. Han, M. Chen, D. Chen, and Y. Li, "Agentic Data Engineering for LLMs: System Design and a Controlled Empirical Study," in *Workshop for Autonomous Machine Learning Research at NeurIPS*, 2026. arXiv:2610.14238 [paper] [code] [homepage]
2. [AAAI 2026] **R. Ding**, W. Cheng, M. Shao, and C. Zhao, "SkillGen: Learning Domain Skills for In-Context Sequential Decision Making," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 **(Oral)** [paper] [code]
3. [ACL 2024] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, W. Zhang, S. Qin, S. Rajmohan, Q. Lin, and D. Zhang, "Everything of Thoughts: Defying the Law of Penrose Triangle for Thought Generation," in *Findings of the Association for Computational Linguistics (ACL)*, 2024 [paper] [code]

Preference Learning & Elicitation

4. [Preprint] [FLLMPT Workshop @NeurIPS 2026] **R. Ding***, T. Gao*, L. Geng, M. Bansal, and Z. Deng, "Who Should We Listen to More? Welfare-Aware Preference Acquisition for Pluralistic Alignment," in *Workshop on Foundations of LLM Post-Training in Changing Environments at NeurIPS*, 2026. arXiv:2610.19035 [paper] [code]
5. [ICML 2026] **R. Ding***, T. Gao*, T. P. Zollo, E. Bachmat, R. Zemel, and Z. Deng, "Whom to Query for What: Adaptive Group Elicitation via Multi-Turn LLM Interactions," in *The 43rd International Conference on Machine Learning (ICML 2026)*, 2026 [paper] [code]

Reliable LLM Evaluation

6. [Preprint] [TAE Workshop @NeurIPS 2026] Q. Shen, X. Li, **R. Ding**, Y. Chen, D. Chen, and Y. Li, "Speculative Evaluation of Stochastic LLMs," in *Can We Trust AI Evaluation? Workshop at NeurIPS*, 2026. arXiv:2609.28560 [paper] [code]
7. [COLM 2026] **R. Ding***, Y. Pang*, H. Sun, Y. Wang, Z. S. Wu, and Z. Deng, "Rubrics as an Attack Surface: Stealthy Preference Drift in LLM Judges," in *Proceedings of the 3rd Conference on Language Modeling (COLM)*, 2026 [paper] [code]
8. [NeurIPS 2024] R. Yang, **R. Ding**, Y. Lin, H. Zhang, and T. Zhang, "Regularizing Hidden States Enables Learning Generalizable Reward Model for LLMs," in *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024 [paper] [code]

Reliable Machine Learning & Applications

9. [SDM 2025] **R. Ding**, X. Zhao, C. Zhao, M. Shao, Z. Chen, and H. Chen, "Evidence-Based Out-of-Distribution Detection on Multi-Label Graphs," in *Proceedings of the 2025 SIAM International Conference on Data Mining (SDM)*, pp. 588–597, SIAM, 2025 [paper]
10. [FSE 2023] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, X. Wu, M. Zhang, Q. Chen, X. Gao, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, "TraceDiag: Adaptive, Interpretable, and Efficient Root Cause Analysis on Large-Scale Microservice Systems," in *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023 [paper] [slides]
11. [KDD 2023] L. Wang, C. Zhang, **R. Ding**, Y. Xu, Q. Chen, W. Zou, Q. Chen, M. Zhang, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, "Root Cause Analysis for Microservice Systems via Hierarchical Reinforcement Learning from Human Feedback," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023 [paper]
12. [VLDB 2023] Y. Chen, C. Zhang, M. Ma, Y. Liu, **R. Ding**, B. Li, S. He, S. Rajmohan, Q. Lin, and D. Zhang, "ImDiffusion: Imputed Diffusion Models for Multivariate Time Series Anomaly Detection," in *Proceedings of the VLDB Endowment (VLDB)*, 2023 [paper] [code]

RESEARCH EXPERIENCE

Research Assistant, **University of North Carolina at Chapel Hill** Fall 2025 – Present
Chapel Hill, NC | Advised by Dr. Zhun Deng.

- **Welfare-Aware Preference Acquisition (WAPA)**. Allocates limited human feedback across individuals using a social welfare objective, improving preference supervision while accounting for underrepresented users and group priorities. *(Preprint; accepted at NeurIPS 2026 FLLMPT Workshop.)*
- **Group-Adaptive Elicitation (GAE)**. Jointly selects whom to query and what to ask in multi-turn LLM interactions. Combines information-gain-based question selection with graph-based response imputation to infer population preferences under limited budgets. *(Accepted at ICML 2026.)*
- **Rubrics as an Attack Surface**. Investigates how subtle edits to evaluation rubrics systematically shift LLM judges' preferences while passing benchmark validation, exposing a vulnerability in evaluation and alignment pipelines. *(Accepted at COLM 2026.)*

Research Intern, **Alibaba Token Foundry (Tongyi Lab)** May 2026 – Aug 2026
Hangzhou, China | Advised by Dr. Qianli Shen and Daoyuan Chen.

- **Agentic Data Engineering (ADE)**. Enables LLM agents to iteratively design and refine data-selection, reward, and curriculum strategies using experimental feedback and accumulated findings. Evaluates whether these strategies improve target models on held-out benchmarks under controlled training conditions. *(Preprint; accepted at NeurIPS 2026 AutoMLR Workshop.)*
- **Speculative Evaluation (SpecEval)**. Uses hierarchical Bayesian estimates to allocate rollout budgets across benchmark tasks, reducing uncertainty in stochastic LLM evaluation. Speculative asynchronous execution reduces the synchronization overhead of pilot sampling. *(Preprint; accepted at NeurIPS 2026 TAE Workshop.)*

Research Intern, **Microsoft Research** May 2024 – Aug 2024
Redmond, WA | Advised by Dr. Minghua Ma and Dr. Ze Li.

- **Triangle: Incident Triage with Multi-LLM-Agents**. Coordinates specialized LLM agents to interpret cloud incidents, collect troubleshooting evidence, and negotiate assignment to responsible teams, automating incident triage across distributed team knowledge.

Research Assistant, **University of Illinois Urbana-Champaign** Mar 2024 – Jul 2024
Urbana, IL | Advised by Dr. Tong Zhang, working with Rui Yang.

- **Generalizable Reward Model (GRM)**. Regularizes reward-model hidden states with an auxiliary language-modeling objective to improve generalization to unseen prompts and responses and mitigate reward over-optimization in RLHF. *(Accepted at NeurIPS 2024.)*

Research Intern, **Microsoft Research Asia** Nov 2022 – Aug 2023
Beijing, China | Advised by Dr. Lu Wang and Dr. Chaoyun Zhang.

- **Everything of Thoughts (XoT)**. Combines Monte Carlo tree search with LLM-guided thought generation and revision to incorporate domain knowledge into reasoning and explore flexible solution structures. *(Accepted at Findings of ACL 2024.)*
- **TraceDiag**. Uses reinforcement learning to prune distributed trace graphs and identify root causes in large-scale microservice systems, making diagnosis more efficient and interpretable. *(Accepted at ESEC/FSE 2023.)*
- **Root Cause Analysis via RLHF**. Incorporates expert feedback into hierarchical reinforcement learning for causal discovery and root cause localization in microservice systems. *(Accepted at KDD 2023.)*

Research Assistant, **Tianjin University** Aug 2021 – Aug 2022
Tianjin, China | Advised by Dr. Minglai Shao and Dr. Wenjun Wang; collaborated with Dr. Xujiang Zhao and Dr. Chen Zhao.

- **Multi-Label Out-of-Distribution Detection**. Uses evidential graph neural networks to quantify label uncertainty and identify nodes with unseen labels, extending graph OOD detection to multi-label settings. *(Accepted at AAAI 2023 UDM Workshop.)*
- **Temporal Community Structure**. Combines variational network embeddings, community priors, and recurrent modeling to discover communities and track their evolution in dynamic networks. *(Published in IEEE Transactions on Cybernetics, 2023.)*

HONORS AND FELLOWSHIPS

Doctoral Merit Fellowship, University of North Carolina at Chapel Hill 2025–2026

Merit Scholarship, Georgia Institute of Technology 2022–2023

ACADEMIC SERVICE

- Conference reviewer: AAAI 2026–2027, ICLR 2026–2027, COLM 2026, NeurIPS 2026, ICML 2026, KDD 2025–2026.